

State-Space Sequence Models for Non-Invasive Brain-to-Text Decoding: A Hybrid Mamba-2 Architecture Trained on the SpanishBCBL MEG Dataset

COM865 Research Project Proposal — Extending Brain2Qwerty with Selective State-Space Sequence Modelling

August 2026 — built on a verified reproduction of the published V1 baseline (test CER 0.389, 4× V100, Kelvin2 HPC)

1. Introduction and Motivation

Decoding produced language from non-invasive brain recordings is one of the central open problems in applied neuroengineering: it promises communication neuroprostheses without surgical risk, but non-invasive signals sit orders of magnitude below invasive recordings in signal-to-noise ratio. The Brain2Qwerty study¹ recently demonstrated that this gap is narrower than previously assumed: a convolution-plus-transformer model trained on MEG recordings of participants typing memorised sentences reaches a character error rate (CER) of 32% with language-model rescoring (~38% without it), and as low as 19% for the best participants. The dataset underlying that result — SpanishBCBL, 35 native Spanish speakers typing 5–8 word sentences on an MR-compatible QWERTY keyboard, ~5.1K MEG sentences / ~193K characters — is publicly available, and Meta has released two reference codebases: V1 (per-keystroke classification) and V2 (whole-sentence CTC transcription with contrastive and LLM-based stages, released for the companion English cohort).

Both reference architectures share one structural choice that this proposal challenges: the sequence-modelling core is built entirely from attention (a sentence-level transformer in V1, a Conformer in V2). Attention is a questionable inductive bias for typing signals. Keystroke-related motor events occur at irregular, subject-specific intervals; the informative structure is a smooth, event-driven dynamical process rather than an all-pairs token interaction; and attention's quadratic cost caps the temporal context that can be modelled at a fixed compute budget. Selective state-space models — Mamba and its second-generation formulation based on structured state-space duality (SSD)^{9,10} — offer precisely the complementary bias: linear-time sequence modelling with input-dependent state transitions that natively emphasise or skip time steps according to the signal. This proposal introduces a hybrid Mamba-2 architecture for MEG keystroke decoding, trained on SpanishBCBL, that replaces the attention core with interleaved Mamba-2 blocks and sparse global-attention layers while retaining the strongest validated components of the V2 training framework.

The project stands on a verified foundation: the V1 baseline has been reproduced end-to-end on the Kelvin2 HPC cluster (feature cache over 81 usable recordings, DDP training on 4× V100), reaching test CER 0.389 against the paper's ~0.38 for the same configuration (Conv+Transformer, no language model), with best validation CER 0.381. Every number proposed below is therefore anchored to a working pipeline, not a paper plan.

2. State of the Art: Decoding Language from MEG

2.1 From perception to production decoding

Modern non-invasive language decoding began with perception. Défossez et al.² showed that perceived speech segments can be retrieved from 3-second MEG/EEG windows with a contrastive model aligned to wav2vec 2.0 representations (up to 41% top-10 accuracy from MEG), introducing two ideas that now pervade the field: per-subject input layers and contrastive alignment with pretrained speech/language models. Tang et al.³ demonstrated semantic reconstruction of perceived and imagined language from fMRI using a language-model decoder, establishing the neural-encoder-plus-language-model template. Decoding production is harder and younger: before Brain2Qwerty¹, the strongest non-invasive production result decoded only 10 letters from EEG at 75.8% CER (Crell and Müller-Putz, 2024), while Brain2Qwerty's 32% CER over a 29-character alphabet on MEG — against 67% on EEG for the identical model — remains the reference point and quantifies why this proposal works on MEG rather than EEG.

2.2 Speech-recognition-style MEG decoders

A second line of work treats MEG as a distorted speech signal and imports automatic-speech-recognition machinery wholesale. NeuSpeech and its successor MAD (Yang et al., 2024) transform raw MEG and feed it into a pretrained Whisper model for open-vocabulary text decoding without teacher forcing, reporting state-of-the-art MEG-to-text results on perception tasks. BrainECHO (2024)⁴ inserts a vector-quantised spectrogram autoencoder between brain signals and Whisper to improve generalisation. In the EEG literature, EEG-to-Text (Wang and Ji, 2022) and DeWave (Duan et al., 2023) align word-level EEG embeddings with BART via a pretrain-finetune paradigm. Meta's V2 codebase internalises the same lesson for production decoding: char-level CTC (Graves et al.¹³) for alignment-free training, a Conformer sequence core¹², a word-level SigLIP contrastive loss matched to frozen LLM token embeddings via dynamic time warping, and a LoRA-adapted LLM decoding stage — a full ASR stack transplanted onto brain signals. The proposed architecture inherits this framing.

2.3 Benchmarks and the scaling signal

The 2025 PNPL competition^{5,6} (LibriBrain: >50 hours of within-subject MEG during audiobook listening, with speech-detection and phoneme-classification leaderboards) has given the field its first ImageNet-style yardstick and produced two findings directly relevant here. First, within-subject data depth drives decoding gains more than cohort breadth — supporting a design that extracts maximal per-sentence context. Second, competition-winning solutions reformulate classification as sequence reconstruction with auxiliary speech-feature targets, echoing V2's multi-objective design and validating the staged, auxiliary-loss training regime adopted below.

2.4 State-space models for neural signals

Structured state-space sequence models (S4) and their selective successor Mamba⁹ model long sequences in linear time with input-dependent dynamics; Mamba-2's SSD formulation¹⁰ unifies SSMs and attention and enables larger state dimensions at transformer-competitive quality. In the biosignal domain the evidence is accumulating rapidly: EEGMamba (Wang et al., 2025) uses Mamba blocks as the backbone of a multi-task EEG foundation model; MI-Mamba (2025) shows a

CNN+Mamba hybrid beating prior state of the art on motor-imagery classification with roughly 6× fewer parameters; bidirectional Mamba variants (BiMamba, EmotionMamba) improve long-sequence EEG emotion recognition; and State Mamba (Weng et al., AAAI 2026) models spatiotemporal state transitions explicitly, topping transformer-based EEG foundation models on three public benchmarks. Closest to this proposal, a NeurIPS 2025 study demonstrates generalizable, real-time neural decoding with hybrid state-space models — evidence that SSM-attention hybrids are viable specifically for neural decoding, not only for classification benchmarks. In large-scale language modelling, NVIDIA's Nemotron-H family¹¹ established the hybrid pattern this proposal borrows: mostly Mamba-2 layers with a small fraction of interleaved self-attention layers recovering exact long-range retrieval at a fraction of the inference cost of a pure transformer.

2.5 The gap this project occupies

Three facts define the opening. (i) Production decoding from MEG is under-explored relative to perception — essentially one published model family (Brain2Qwerty V1/V2), both attention-based. (ii) SSMS have proven themselves on EEG classification and foundation-model benchmarks but have never been applied to sentence-level brain-to-text decoding from MEG. (iii) The V2 training framework (CTC + word-contrastive + LLM decoding) has not been evaluated on SpanishBCBL at all — it was released for the English cohort. This project sits at the intersection: the first state-space sequence model for MEG production decoding, trained with the strongest available framework, on the dataset for which a verified reproduction baseline already exists.

3. Proposed Methodology

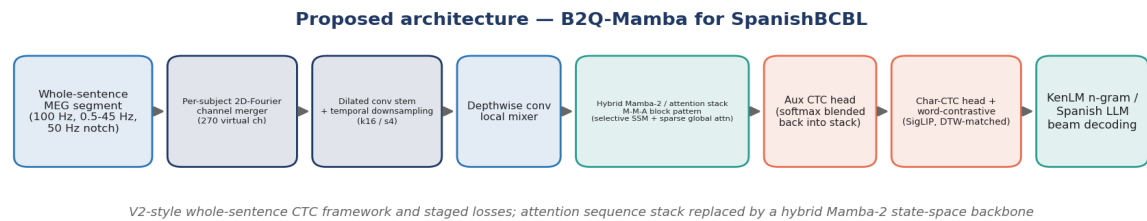


Figure 1 — Proposed B2Q-Mamba architecture. Grey/blue components are inherited from the validated V1/V2 pipelines; the Mamba-2 hybrid stack is the new contribution.

3.1 Data pipeline and preprocessing (from V2)

The unit of modelling is the whole typed sentence, not the individual keystroke. Continuous MEG is resampled to 100 Hz, bandpass-filtered at 0.5–45 Hz with a 50 Hz notch, RobustScaler-normalised, and segmented per sentence. This departs deliberately from V1's 0.5 s per-keystroke windows (50 Hz, 0.1–20 Hz): the wider band retains beta/low-gamma motor content that V1 discards by construction, and whole-sentence segments let the sequence model — rather than the label timestamps — decide where keystroke information lives. V2's on-device augmentation suite is applied during training only: per-channel constant offsets (SD 0.3) simulating session drift, SpecAugment time/frequency masking, and time-stretch, the last being particularly apt for typing data since it varies inter-keystroke rhythm without changing the character sequence. With only 81 usable recordings, this augmentation is a necessity, not a luxury.

3.2 Front-end: subject-conditioned convolutional encoder

The front-end retains the component both released versions agree on — the per-subject 2D-Fourier channel merger projecting the sensor array onto 270 learned virtual channels — followed by a dilated convolutional stem and learned temporal downsampling (kernel 16, stride 4) that compresses the waveform into a frame sequence at ~ 6 Hz. Per-subject conditioning matters here because the reproduction confirmed subject identity as the dominant error source (per-subject CER spanning 0.287–0.552 on the baseline); the merger is the architecture's primary mechanism for absorbing inter-subject anatomical and session differences before sequence modelling begins.

3.3 Core contribution: hybrid Mamba-2 sequence stack

The Conformer/transformer core is replaced by a hybrid stack in the Nemotron-H pattern¹¹: groups of three Mamba-2 blocks followed by one global self-attention layer (M-M-A), repeated twice, with a depthwise convolution as the local mixer in front of the stack (preserving the local smoothing that both V1's conv encoder and V2's Conformer rely on). Each Mamba-2 block uses the SSD parametrisation — d_{model} matched to the encoder output, state dimension 64–128, head dimension 64, expansion factor 2 — implemented with the reference mamba-ssm kernels. The design rests on three methodological arguments:

Inductive bias. Typing is an event-driven process: motor preparation, keypress, and inter-key intervals of irregular duration. Mamba's selective scan makes the state transition input-dependent, so the model can hold motor state across a long pause and update sharply at a keypress — the continuous-time analogue of the task's dynamics. Attention must learn this behaviour from scratch through positional encodings that assume a uniform clock.

Compute and context. Sequence cost grows linearly in segment length, so the same $4\times V100$ budget that trains V2 on single sentences can train the Mamba variant on longer contexts — multi-sentence spans that let the model exploit cross-sentence language statistics, the single most predictable source of CER reduction given that the reproduction's residual errors are predominantly single-character substitutions in otherwise correct sentences.

Why not pure Mamba. SSMs are known to underperform at exact long-range retrieval — precisely what is needed to resolve an ambiguous character using a word typed two seconds earlier. Interleaving one attention layer per three Mamba blocks restores retrieval at $\sim 25\%$ of the attention compute of a pure transformer, a trade validated at scale by Nemotron-H¹¹ and, for neural signals specifically, by hybrid-SSM real-time decoding work at NeurIPS 2025. The M-M-A ratio is an explicit ablation axis (M-only, M-A, M-M-M-A) against the Conformer core at matched parameter count.

3.4 Training objectives (adopted and adapted from V2)

Training follows V2's staged multi-objective schedule, which the LibriBrain competition results independently support as the right regime for MEG sequence decoding⁵:

Stage	Objective	Rationale for this project
From epoch 0	Character-level CTC on final logits + auxiliary CTC head whose softmax is blended back into the stack input (blend $\alpha=0.7$)	Removes V1's brittle assumption that the logged keypress timestamp is the neural event time; the aux head deepens supervision of the early stack — important because Mamba blocks have no attention shortcut

Stage	Objective	Rationale for this project
Mid-training	Word-level SigLIP contrastive loss: CTC-segmented pseudo-word embeddings, DTW-matched to frozen LLM token embeddings	Injects lexical structure the character CTC cannot represent; targets exactly the substitution-type errors dominating the reproduction's residual error mass
Late stage (optional)	LLM decoding loss: LoRA-adapted Spanish LLM conditioned on CTC greedy text + adapted neural word embeddings	Replaces V1's offline 9-gram rescoring with a learned decoder; kept optional because Spanish LLM selection is a separate decision and KenLM Spanish n-grams provide the primary linguistic baseline

Table 1 — Staged training objectives and their justification.

Optimisation uses V2's settings (AdamW lr 8e-4, weight decay 1e-3, warmup+cosine schedule, bf16 mixed precision, gradient accumulation 2), with one deliberate exception: the Mamba blocks are initialised with a smaller effective learning rate or a short SSM-only warmup, since selective-scan parameters are more sensitive to early-training instability than attention layers — a standard precaution in hybrid SSM training.

4. Evaluation Protocol

Primary metric: sentence-level CER (micro-averaged) on the held-out test split, directly comparable to the reproduced 0.389 baseline and the paper's 0.38/0.32 figures (without/with language model). Secondary metrics: WER, exact-sentence rate, per-subject CER distributions, and the semantic error rate (RoBERTa-large embedding distance) adopted from V2 for the LLM decoding stage. All comparisons use the V1 deterministic subject-disjoint split and identical early stopping (val CER, patience 30). The ablation grid is the scientific core of the project: (a) V2 framework on SpanishBCBL with its original Conformer core — isolates the dataset/framework transfer; (b) the same with the hybrid Mamba-2 core at matched parameters — isolates the architecture variable; (c) M-M-A pattern variants and a bidirectional Mamba variant (motivated by BiMamba results on EEG) — isolates the design choice; (d) decoder study (none / KenLM / LLM) — isolates the linguistic contribution. Each cell is a same-budget training run, so every claim in the final report is a controlled comparison against the verified baseline.

5. Expected Contributions and Risks

Contributions: (1) the first state-space sequence model for sentence-level brain-to-text decoding from MEG production data; (2) the first evaluation of the V2 CTC/contrastive/LLM framework on SpanishBCBL, extending a framework released only for English; (3) a controlled, same-budget comparison of attention, Conformer, and hybrid Mamba-2 cores on an identical pipeline — a result of value to the MEG decoding community whichever way it resolves; (4) an open, reproducible extension of Meta's reference codebase.

Risks and mitigations. The dataset is small (~81 recordings): mitigated by the augmentation suite, per-subject merger, and the staged schedule that delays data-hungry objectives. Mamba blocks are less battle-tested on small neural datasets: mitigated by keeping the Conformer-core run (ablation a) as a guaranteed contribution, and by the hybrid design itself, which degrades gracefully toward a

sparse-attention model. Spanish LLM availability for the decoding stage: mitigated by KenLM Spanish n-grams as the primary linguistic decoder, with the LLM stage optional and clearly marked as such.

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